

Proof. 1. **Degenerate / edge cases.** If $x = 0$ then $\|x + y\| = \|y\| = \|0\| + \|y\|$; similarly if $y = 0$. Hence the inequality holds with equality. From now on assume $x \neq 0$ and $y \neq 0$.

2. **Squaring the inequality.** The norm is non-negative, so $\|x + y\| \leq \|x\| + \|y\|$ is equivalent to

$$\|x + y\|^2 \leq (\|x\| + \|y\|)^2. \quad (1)$$

3. **Expansion of $\|x + y\|^2$.** Using sesquilinearity of the inner product (linearity in the first argument, conjugate-linearity in the second) we have

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle y, y \rangle + \langle x, y \rangle + \langle y, x \rangle \\ &= \|x\|^2 + 2 \operatorname{Re} \langle x, y \rangle + \|y\|^2. \end{aligned} \quad (2)$$

4. **Cauchy–Schwarz inequality.** For any $x, y \in V$,

$$|\langle x, y \rangle| \leq \|x\| \|y\|, \quad (3)$$

with equality iff x and y are linearly dependent. Since $\operatorname{Re} \langle x, y \rangle \leq |\langle x, y \rangle|$, we obtain

$$\operatorname{Re} \langle x, y \rangle \leq \|x\| \|y\|. \quad (4)$$

5. **Substitution into (2).** Inserting the bound (4) into (2) gives

$$\begin{aligned} \|x + y\|^2 &\leq \|x\|^2 + 2\|x\| \|y\| + \|y\|^2 \\ &= (\|x\| + \|y\|)^2. \end{aligned} \quad (5)$$

Inequality (5) is precisely (1).

6. **Taking square roots.** Both sides of (5) are non-negative, therefore

$$\|x + y\| \leq \|x\| + \|y\|.$$

7. **Equality condition.** Equality in the final inequality requires equality in every intermediate step:

- Equality in (4) forces $|\langle x, y \rangle| = \|x\| \|y\|$, which by the equality case of Cauchy–Schwarz means $y = \lambda x$ for some scalar λ .
- Moreover we need $\operatorname{Re} \langle x, y \rangle = |\langle x, y \rangle|$. If $y = \lambda x$ then $\langle x, y \rangle = \lambda \langle x, x \rangle = \lambda \|x\|^2$, so $\operatorname{Re} \langle x, y \rangle = \operatorname{Re}(\lambda) \|x\|^2$ and $|\langle x, y \rangle| = |\lambda| \|x\|^2$. Equality of the real part with the modulus forces λ to be a non-negative real number.

Consequently,

$$\|x+y\| = \|x\|+\|y\| \iff \exists \lambda \geq 0 \text{ such that } y = \lambda x \quad (\text{equivalently } x = \mu y, \mu \geq 0).$$

Thus the triangle inequality holds for all vectors in an inner-product space,
and the equality case has been fully characterized. $\square$

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